FaultGuard AI - Pipeline & Architecture

1. Data Pipeline

Raw data is extracted from multi-sheet Excel files. Features are normalized using StandardScaler and segmented into 200-sample windows with a stride of 50. Label encoding handles 11 distinct fault scenarios.

2. Model Architecture (1D-CNN)

- Backbone: 3x Conv1D layers (64, 128, 256 filters) with BatchNormalization and ReLU.
- Multi-Task Heads:
  a. Detection (Sigmoid): Normal vs Fault.
  b. Classification (Softmax): 11-class fault identification.
  c. Location (Sigmoid + Rescaling): Precise KM regression (0.0 - 5.0 KM).

3. Backend & Deployment

The backend is built with Flask, serving predictions via a REST API. It is containerized using Docker and compatible with Hugging Face Spaces. Frontend is a Nuxt 4 application with Chart.js visualization.

4. Technical Specifications

- System Frequency: 50 Hz
- Transmission Line: 5.0 km
- Voltage Base: 200 kV
- Current Base: 1 kA
- Thresholds: V < 0.85, I > 0.3 p.u.